

# AP Chemistry: The Complete Exam Guide

All 9 units explained, the long and short free-response questions, a calculation playbook, and 10 hard practice questions with full explanations.

Hi, I'm **Pigo!** I'll drop tips all through this guide.

**3:15**

hours total  
90 + 105 minutes

**60+7**

multiple-choice +  
free-response questions

**9 units**

atoms to  
electrochemistry

**75–78%**

scored 3 or higher  
in recent years

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# 01 THE EXAM AT A GLANCE

## Two sections, one score

Your AP score (1–5) combines a multiple-choice section and a free-response section, each worth half. Both mix conceptual explanations with calculations.

**50%**

Section I • Multiple choice • 60 questions

**50%**

Section II • Free response • 7 questions

### Exam structure

| Section             | Weight & timing                             | What you do                                                                                                                                                                      |
|---------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I • Multiple choice | 50% of score<br>1 hr 30 min<br>60 questions | Questions on models, experiments, particle diagrams and calculations. A scientific or graphing calculator is recommended.                                                        |
| II • Free response  | 50% of score<br>1 hr 45 min<br>7 questions  | <b>3 long questions</b> worth 10 points each and <b>4 short questions</b> worth 4 points each. Explain models, design experiments, represent data, calculate and justify claims. |

#### HYBRID DIGITAL

##### Click in Bluebook, write on paper

You answer multiple-choice questions and read the free-response questions in the **Bluebook** app, then handwrite your free-response answers in a paper booklet. Practice writing clear, organized work by hand.

#### PROVIDED ON EXAM DAY

##### Periodic table + equation sheet

A periodic table and a sheet of equations and constants are provided for both sections. You still need to know **which** equation applies and what each symbol means.

### How to use this guide

#### Just starting?

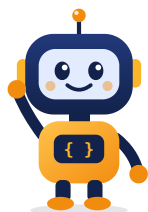
Read pages 2–6, then follow the roadmap on page 8 month by month.

#### Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

#### Exam in weeks?

Take the 10 questions on page 9 in 20 minutes, then use the final-four-weeks plan.



**Pigo's tip:** AP Chem rewards **particle-level explanations**. Practice explaining with intermolecular forces, Coulomb's law and collisions, not just numbers.

## SECTION I

# 02 The multiple-choice section

60 questions in 90 minutes gives you 1 minute 30 seconds per question.  
Calculations are designed to be done quickly, often by estimating.

**60**

questions with four  
answer choices

**1:30**

minutes per question on  
average

**50%**

of your total AP score

**0**

points lost for a wrong  
answer, so never leave  
one blank

### How the questions are weighted by unit

College Board gives a range for each unit's share of the exam. Bars show minimum to maximum.

#### Unit 3

Properties of Substances &  
Mixtures

**18–22%**

#### Unit 8

Acids & Bases

**11–15%**

#### Units 1–2

Atomic & Compound  
Structure (each)

**7–9%**

#### Units 4–7

Reactions, Kinetics, Thermo,  
Equilibrium (each)

**7–9%**

#### Unit 9

Thermodynamics &  
Electrochemistry

**7–9%**

0%

10%

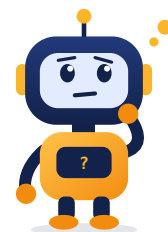
20%

**What this means for your study time:** Unit 3 (intermolecular forces, gases, solutions) and Unit 8 (acids and bases) are the biggest. Unit 3 ideas also explain answers across the rest of the course.

### Strategies that work on this section

- **Estimate before calculating.** Answer choices are often far enough apart that rounding works.
- **Draw particle diagrams** for "what does the mixture look like" questions.
- **Use Coulomb's law reasoning** for ionization energy, lattice energy and atomic size.
- **Check units and sig figs** only after the chemistry is right.
- **Know your strong acids and bases** and the common soluble and insoluble ions.
- **Flag and move on** after 2 minutes. Return once the quick points are banked.

Many calculations are **designed for estimation**. If the numbers look messy, round them!



# 03 WHAT THE COURSE COVERS

## The 9 units, in plain English

Everything on the exam fits into these nine units.

UNIT

1

7–9%

### Atomic Structure & Properties

Moles, electrons and periodic trends.

**Be ready to:** convert moles, mass and particles; read mass and photoelectron spectra; write electron configurations; explain trends with Coulomb's law.

UNIT

2

7–9%

### Compound Structure & Properties

Bonding and molecular shape.

**Be ready to:** compare ionic, covalent and metallic bonding; draw Lewis structures and formal charges; use VSEPR and hybridization; predict polarity.

UNIT

3

18–22%

### Properties of Substances & Mixtures

Intermolecular forces, gases and solutions.

**Be ready to:** rank IMFs and boiling points; use the ideal gas law and kinetic molecular theory; calculate molarity; explain chromatography, distillation and Beer's law.

UNIT

4

7–9%

### Chemical Reactions

Equations, stoichiometry and reaction types.

**Be ready to:** write net ionic equations; do stoichiometry and limiting reactant problems; run titration calculations; identify acid-base and redox reactions.

UNIT

5

7–9%

### Kinetics

How fast reactions happen and why.

**Be ready to:** find rate laws from data; use integrated rate laws and half-life; connect mechanisms to rate laws; explain catalysts and activation energy.

UNIT

6

7–9%

### Thermochemistry

Heat in reactions.

**Be ready to:** use  $q = mc\Delta T$  and calorimetry; apply Hess's law and bond enthalpies; relate  $\Delta H$  to bond breaking and forming.

UNIT

7

7–9%

### Equilibrium

Reversible reactions and their balance point.

**Be ready to:** write  $K$  and  $Q$ ; use ICE tables; apply Le Châtelier's principle; calculate solubility with  $K_{sp}$  and the common-ion effect.

UNIT

8

11–15%

### Acids & Bases

pH, buffers and titrations.

**Be ready to:** calculate pH for strong and weak acids and bases; use Henderson-Hasselbalch for buffers; read titration curves; relate  $pK_a$  to structure.

UNIT

9

7–9%

### Thermodynamics & Electrochemistry

What makes reactions favorable, and batteries.

**Be ready to:** predict signs of  $\Delta S$  and  $\Delta G$ ; relate  $\Delta G^\circ$ ,  $K$  and  $E^\circ$ ; describe galvanic and electrolytic cells; use Faraday's law.

## SECTION II

04

# The 7 free-response questions

105 minutes, 3 long questions (10 points each) and 4 short questions (4 points each). Long questions combine several units around one scenario.

Q1 • 10 pts

Q2 • 10 pts

Q3 • 10 pts

Q4 • 4

Q5 • 4

Q6 • 4

Q7 • 4

46 raw points in total, worth 50% of your AP score

**Long****Three 10-point questions**

Q1–Q3 • about 23 minutes each

One scenario, many parts: calculations, particle-level explanations, experimental design and data analysis, often across several units.

Later parts may use earlier answers. If you are stuck, assume a value and keep going.

**Short****Four 4-point questions**

Q4–Q7 • about 9 minutes each

A single focused task: one calculation, one explanation, or one representation such as a particle diagram or Lewis structure.

Short questions are quick points. Don't leave them to the end.

**Calc****Calculation parts**

Show the setup

Show the equation, substitute with units, and give an answer with appropriate significant figures.

A correct setup with an arithmetic slip can still earn most of the credit.

**Why****Explanation parts**

Justify with particles

Explain using intermolecular forces, Coulombic attraction, collisions, or comparisons of Q and K.

Compare both substances. "A has stronger IMFs than B because...".

## ◆ A time plan for the 105 minutes

Q1

~23 min

Q2

~23 min

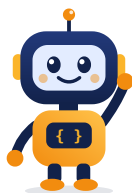
Q3

~23 min

Q4–Q7

~9 min each

A suggested split, not a rule. Start with the question type you're strongest at, and never leave a question blank.



Show **every step** of a calculation. Most points are for the setup, not the final number.

FREE-RESPONSE PLAYBOOK

05

# Anatomy of a full-credit answer

Here is a short buffer prompt with a full-credit answer, and the habits that earn points.

PRACTICE TASK • WRITTEN BY ACT SAT GO

**Practice prompt (short-question style):** A buffer contains 0.20 M  $\text{CH}_3\text{COOH}$  and 0.30 M  $\text{CH}_3\text{COONa}$ . The  $\text{pK}_a$  of acetic acid is 4.74. (a) Calculate the pH. (b) A small amount of HCl is added. Write the net ionic equation for the reaction that keeps the pH nearly constant. (c) Explain whether the pH increases, decreases or stays the same.

Full-credit response

① (a)  $\text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = 4.74 + \log(0.30/0.20) = 4.74 + 0.18 = \mathbf{4.92}$

② (b)  $\text{H}_3\text{O}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq}) \rightarrow \text{CH}_3\text{COOH}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

③ (c) The pH **decreases slightly**. Added  $\text{H}_3\text{O}^+$  converts some  $\text{CH}_3\text{COO}^-$  into  $\text{CH}_3\text{COOH}$ , so  $[\text{A}^-]/[\text{HA}]$  becomes smaller.

④ Because both concentrations are large compared with the added acid, the ratio and the pH change only a little.

- ① Equation, substitution, answer, all shown.
- ② Net ionic equation: only the species that react; spectator ions left out.
- ③ Direction + reason from the change in the ratio.
- ④ Explain the "buffer" part: why the change is small.

## Five habits for every free-response question

1

Answer every lettered part, in order, with a label.

2

Show the equation, the substitution and the units.

3

Justify with a cause → effect sentence, not a word.

4

Use the reference sheet; you don't need to memorize it.

5

Blank part? Write the relationship you would use. It can earn credit.

# 06 LEARN FROM OTHERS

## 10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.

Breaking bonds **releases** energy! (Nope.)



### 01 Bond breaking releases energy

Breaking bonds always **absorbs** energy. Forming bonds releases it.

### 02 IMFs vs bonds

Boiling breaks intermolecular forces, not covalent bonds within molecules.

### 03 Spectator ions in net ionic

Strong electrolytes in solution appear as ions; cancel spectators; keep solids, liquids, gases and weak acids whole.

### 04 Catalysts and equilibrium

A catalyst speeds up both directions equally. It does not change K or the equilibrium amounts.

### 05 Temperature and K

Only temperature changes K. Concentration and pressure changes shift the position, not K.

### 06 Weak vs strong

A dilute strong acid is still strong; a concentrated weak acid is still weak. Strength means extent of ionization.

### 07 Units in the gas law

Use kelvin, and match R to your pressure and volume units.

### 08 Rate law from the equation

Rate laws come from experiments or the slow step of a mechanism, not from the balanced equation's coefficients.

### 09 Signs of $\Delta G$ , $E^\circ$ and K

Favorable:  $\Delta G^\circ < 0$ ,  $E^\circ > 0$ ,  $K > 1$ . Keep all three consistent.

### 10 One-sided comparisons

"A has strong IMFs" isn't enough. Compare: "A has stronger IMFs than B because...".

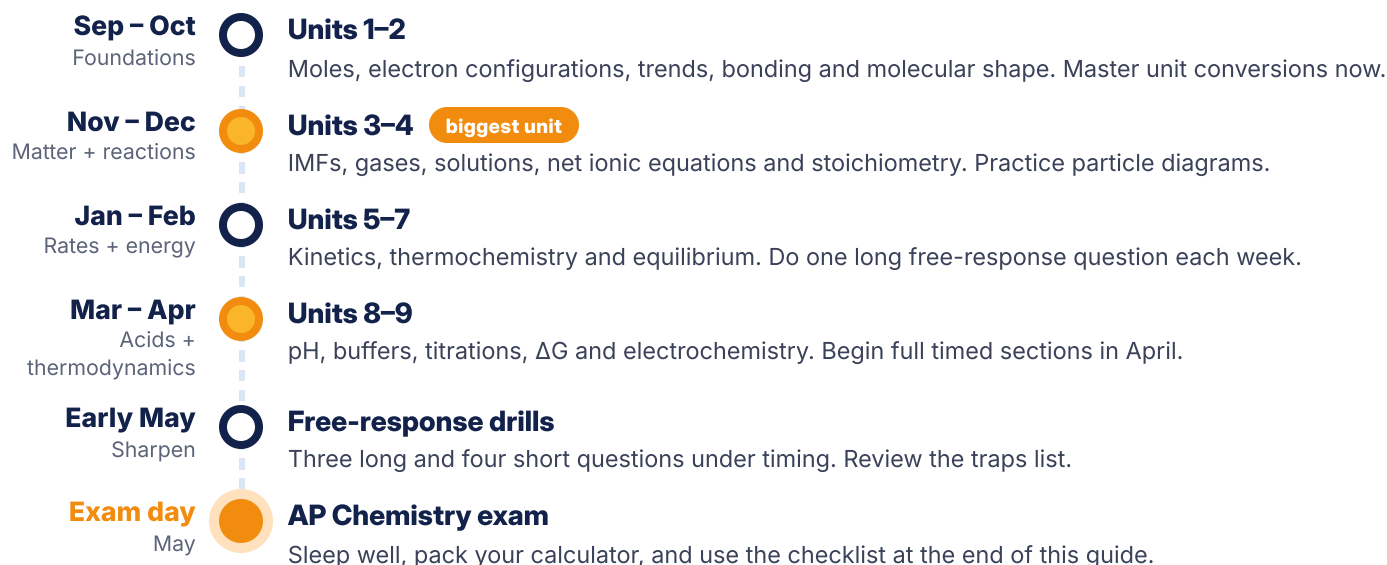
**Quick self-check:** why does HF boil at a higher temperature than HCl, even though HCl has more electrons? (HF forms hydrogen bonds, which are stronger than the extra dispersion forces in HCl.)

07

PLAN YOUR YEAR

# Your study roadmap to May

A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, and keep practicing calculations every week.



## ◆ The final four weeks

| Week        | Focus                                                                       | Target                        |
|-------------|-----------------------------------------------------------------------------|-------------------------------|
| 4 weeks out | Diagnose: 30 multiple-choice questions plus one long free-response question | List your three weakest units |
| 3 weeks out | Fix weak units; two short free-response questions daily                     | 80%+ on unit drills           |
| 2 weeks out | Full 3-hour-15-minute practice exam                                         | Finish both sections on time  |
| Final week  | Light review: net ionic rules, IMFs, equilibrium logic                      | Rest the night before         |

# 08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Give yourself **20 minutes** for all ten, then check page 10.

## 1 UNIT 8 • WEAK ACIDS

What is the pH of 0.10 M acetic acid ( $K_a = 1.8 \times 10^{-5}$ )?

- A 1.00                      B 2.87  
C 4.74                      D 5.74

## 2 UNIT 8 • BUFFERS

A buffer is 0.20 M  $\text{CH}_3\text{COOH}$  and 0.30 M  $\text{CH}_3\text{COONa}$  ( $pK_a = 4.74$ ). What is its pH?

- A 4.56                      B 4.74  
C 4.92                      D 5.04

## 3 UNIT 7 • EQUILIBRIUM

For  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ , equilibrium concentrations are  $[\text{N}_2] = 0.20 \text{ M}$ ,  $[\text{H}_2] = 0.10 \text{ M}$ ,  $[\text{NH}_3] = 0.40 \text{ M}$ . What is  $K_c$ ?

- A 0.80                      B 20  
C 80                        D 800

## 4 UNIT 7 • LE CHÂTELIER

For the exothermic reaction  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ , which change increases the **equilibrium constant**?

- A Lowering the temperature  
B Adding more  $\text{O}_2$   
C Increasing the pressure  
D Adding a catalyst

## 5 UNIT 9 • ELECTROCHEMISTRY

A galvanic cell has  $E^\circ_{\text{cell}} = +1.10 \text{ V}$  and transfers 2 mol of electrons per mole of reaction ( $F = 96,485 \text{ C/mol}$ ). What is  $\Delta G^\circ$ ?

- A  $-106 \text{ kJ/mol}$                       B  $+212 \text{ kJ/mol}$   
C  $-424 \text{ kJ/mol}$                       D  $-212 \text{ kJ/mol}$

## 6 UNIT 9 • ELECTROLYSIS

A current of 2.0 A passes through  $\text{CuSO}_4$  solution for 1,930 s. What mass of Cu (molar mass  $63.55 \text{ g/mol}$ ) is deposited?

- A 0.64 g                      B 1.27 g  
C 2.54 g                      D 5.08 g

## 7 UNIT 5 • KINETICS

Data for  $\text{A} + \text{B} \rightarrow \text{C}$ :

| Trial | [A] (M) | [B] (M) | Rate (M/s)           |
|-------|---------|---------|----------------------|
| 1     | 0.10    | 0.10    | $2.0 \times 10^{-3}$ |
| 2     | 0.20    | 0.10    | $8.0 \times 10^{-3}$ |
| 3     | 0.10    | 0.20    | $2.0 \times 10^{-3}$ |

What is the rate law?

- A  $\text{rate} = k[\text{A}]^2$                       B  $\text{rate} = k[\text{A}]$   
C  $\text{rate} = k[\text{A}][\text{B}]$                       D  $\text{rate} = k[\text{A}]^2 [\text{B}]$

## 8 UNIT 5 • FIRST-ORDER DECAY

A first-order reaction has  $k = 0.0231 \text{ min}^{-1}$ . How long does it take for 75% of the reactant to be consumed?

- A 30 min                      B 45 min  
C 60 min                      D 75 min

## 9 UNIT 3 • IMFS

Which list ranks the substances in order of **increasing** boiling point?

- A  $\text{HCl} < \text{CH}_4 < \text{NaCl} < \text{H}_2\text{O}$   
B  $\text{CH}_4 < \text{H}_2\text{O} < \text{HCl} < \text{NaCl}$   
C  $\text{CH}_4 < \text{HCl} < \text{H}_2\text{O} < \text{NaCl}$   
D  $\text{NaCl} < \text{H}_2\text{O} < \text{HCl} < \text{CH}_4$

## 10 UNIT 9 • THERMODYNAMICS

A reaction has  $\Delta H^\circ = +50 \text{ kJ/mol}$  and  $\Delta S^\circ = +200 \text{ J/(mol}\cdot\text{K)}$ . At what temperatures is it thermodynamically favorable?

- A At all temperatures  
B Only above 250 K  
C Only below 250 K  
D At no temperature

09

CHECK YOUR WORK

# Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

| #  | Answer          | Why                                                                                                                                                                                                    |
|----|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | B • 2.87        | $[H^+] \approx \sqrt{(K_a \cdot C)} = \sqrt{(1.8 \times 10^{-6})} \approx 1.34 \times 10^{-3} \text{ M}$ , so $\text{pH} \approx 2.87$ . Choice A treats it as a strong acid; C is the $\text{pK}_a$ . |
| 2  | C • 4.92        | Henderson-Hasselbalch: $4.74 + \log(0.30/0.20) = 4.74 + 0.18 = 4.92$ . More base than acid, so $\text{pH} > \text{pK}_a$ .                                                                             |
| 3  | D • 800         | $K_c = [\text{NH}_3]^2/([\text{N}_2][\text{H}_2]^3) = 0.16/(0.20 \times 0.001) = 800$ . Choice C forgets to cube $[\text{H}_2]$ .                                                                      |
| 4  | A               | Only temperature changes $K$ . For an exothermic reaction, lowering $T$ shifts products and increases $K$ . Adding $\text{O}_2$ or pressure shifts position only; catalysts change neither.            |
| 5  | D • -212 kJ/mol | $\Delta G^\circ = -nFE^\circ = -(2)(96,485)(1.10) \approx -212,000 \text{ J} = -212 \text{ kJ/mol}$ . Negative, matching a positive $E^\circ$ .                                                        |
| 6  | B • 1.27 g      | Charge = $2.0 \times 1,930 = 3,860 \text{ C} \rightarrow 0.040 \text{ mol e}^-$ . $\text{Cu}^{2+}$ needs $2 \text{ e}^-$ , so $0.020 \text{ mol Cu} \times 63.55 = 1.27 \text{ g}$ .                   |
| 7  | A               | Doubling $[\text{A}]$ quadruples the rate (second order in A). Doubling $[\text{B}]$ changes nothing (zero order in B).                                                                                |
| 8  | C • 60 min      | $t_{1/2} = 0.693/0.0231 = 30 \text{ min}$ . 75% consumed means 25% left: two half-lives = 60 min.                                                                                                      |
| 9  | C               | $\text{CH}_4$ (only dispersion) < $\text{HCl}$ (dipole) < $\text{H}_2\text{O}$ (hydrogen bonding) < $\text{NaCl}$ (ionic lattice).                                                                     |
| 10 | B               | $\Delta G = \Delta H - T\Delta S < 0$ when $T > \Delta H/\Delta S = 50,000/200 = 250 \text{ K}$ . Convert J to kJ consistently.                                                                        |

## 9–10

Strong. Move on to full timed free-response practice.

## 6–8

On track. Revisit the unit behind every miss on page 4.

## 0–5

Build foundations now. Re-learn moles and stoichiometry, then intermolecular forces (Units 1, 3 and 4).

## 10 KNOW THE TARGET

# How students score

In recent years, about 75–78% of students earned a 3 or higher on AP Chemistry, and about 15–18% earned a 5. Steady preparation is what moves a score up a full point.

**75–78%**

of students earned a 3 or higher in recent years

**15–18%**

earned the top score of 5 in recent years

**3+**

is the score many colleges accept for credit or placement

### What each AP score means

**5**

Extremely well qualified

**4**

Very well qualified

**3**

Qualified

**2**

Possibly qualified

**1**

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

### Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, plus **two pencils and a dark-ink pen** for the free-response booklet
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ A **scientific or graphing calculator** with fresh batteries
- ☐ Plan your time: about **90 s** per multiple-choice question; **~23 min** per long and **~9 min** per short free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ Use the provided **periodic table and equation sheet**

**In the free-response section:** show the equation and substitution for every calculation, include units and sensible significant figures, and explain with particle-level reasoning that compares the substances.



ACT SAT GO • 1-ON-1 AP TUTORING

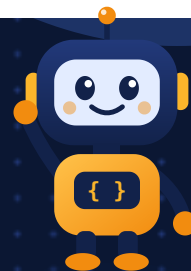
# Aiming for a 5? Get a personal AP Chem plan.

Book a free diagnostic lesson. We'll find your weakest units, review your practice responses, and give you a week-by-week plan to exam day in May.

**Book a free diagnostic lesson →**



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## Concept tutoring

Live 1-on-1 sessions on all nine units, with calculation drills after every lesson.

## Free-response coaching

Practice long and short questions and get feedback against rubric-style criteria.

## Timed exam practice

Full-length multiple-choice sets with score reports, so you see progress before May.

## ◆ Contact us

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